Verifier-Guided Twelve-Tone Composition: A Generate–Verify–Repair Harness for Symbolic Music Generation

Congren Dai[1,2], **Danni Zhao**[1], **Enyang Liu**[1], **Michael Ching Yam**[1], **Zhancheng Guo**[1,2], **Siyi Gu**[1], **Wentao Yang**[1], **Bo Dai**[1], **Xiaobing Li**[1], and **Maosong Sun**[1,2✉]

[1] Central Conservatory of Music [2] Tsinghua University

congren.dai@mail.ccom.edu.cn sms@tsinghua.edu.cn

Abstract

Large language models can produce superficially legal twelve-tone scores that collapse into degenerate textures. We introduce a neuro-symbolic harness that wraps a language-model proposer in a *generate–verify–repair–trace* loop with symbolic verification. The complete pipeline improves event-local consistency without claiming whole-piece legality. Across 40 controlled tasks and four paired models, constraint-checked delivery rises from 13.3% under raw generation to 48.1% with the harness; it abstains on the remaining 51.9% runs. The pass rate of a narrower collision and serialisation-consistency check rises from 33.5% to 58.3%, while degeneracy remains near 0.05, including under adversarial prompting. A blinded evaluation by five experts also shows a descriptive aggregate preference for harness candidates over raw generation in adherence, perceived legality, coherence, and overall quality.

1 Introduction

Twelve-tone (serial) music, a compositional method in which an ordered *tone row* of pitch classes governs harmony and melody through transposition, inversion, and retrograde (Schoenberg 1950; Straus 2016), is an attractive testbed for controllable generation: its core rules are crisp and formal, yet musically interesting serial realisations often depend on global organisation (Perle 1991). More generally, maintaining long-range coherence is a central challenge in symbolic music generation (Huang et al. 2019). This combination exposes a failure mode that is easy to miss when "rule legality" is the only metric. A model instructed to prioritise legality can satisfy the letter of the rules with a *degenerate* score (a few sustained chords, collapsed voices, near-empty bars) that is nominally compliant but musically empty. This behaviour is analogous to *specification gaming*: it satisfies the stated rule-based proxy while defeating the intended musical objective. Reward-hacking work studies the related, but not identical, case in which optimisation exploits a misspecified objective or reward proxy (Amodei et al. 2016; Pan, Bhatia, and Steinhardt 2022; Skalse et al. 2022); here we elicit shortcuts through adversarial prompts rather than training-time reward optimisation.

We argue that a structural intervention can improve constraint satisfaction over prompting alone. A deterministic verifier records hard-rule violations, and a row-aware repair operator projects proposed material back onto the active row slice whenever possible. The large language model (LLM) is retained where it is genuinely useful (high-level planning and note-level proposals that drive musical quality), while symbolic components control pitch-class assignment and surface unresolved violations explicitly. Unlike prompt-only self-correction, the symbolic layer evaluates executable events rather than model-generated claims. Its trace records which row positions survive repair, making dropped or modified material inspectable in the delivered artefact.

This setting is a clean microcosm of a broader challenge in neuro-symbolic generation: automatic proxies can be satisfied without producing useful outputs, and process-level checks need to be distinguished from independent checks on the delivered artefact. Twelve-tone music sharpens the tension because legality is a crisp Boolean property, whereas musical quality is global and is precisely what a legality-maximiser sacrifices. Practically, the harness returns either a candidate passing a final constraint check or an explicit failure with an inspectable record of retained, repaired, and dropped material. Methodologically, separating process diagnostics from retained-artefact checks offers a useful pattern for other structured generators, such as code or plans, even though their domain predicates would need to be redesigned. The target rule set is also unusually hard to enforce by decoding: it includes cross-voice and long-range properties (each voice must track a scheduled twelve-tone stream, and no two simultaneously sounding voices may share a pitch class), so it cannot be checked from a single token prefix. To study it we run six language models across 40 controlled tasks with length and texture sweeps, adversarial (red-team) briefs, module ablations, a corpus-distance analysis against 20 human serial works, and a blind pairwise study with five experts.

Concretely, we (1) formalise serial composition as constrained sequence generation over $\mathbb{Z}_{12}$: an order-48 transformation group, a core event-consistency target defined by hard constraint predicates, and separate process diagnostics and final-check decisions (Section 3); (2) introduce a *generate–verify–repair–trace* harness with a row-preserving repair operator and a generator/verifier *lockstep* invariant, yielding event-local row projection and monotone best-attempt selection rather than unconditional whole-piece legality (Sec-

✉Corresponding author.

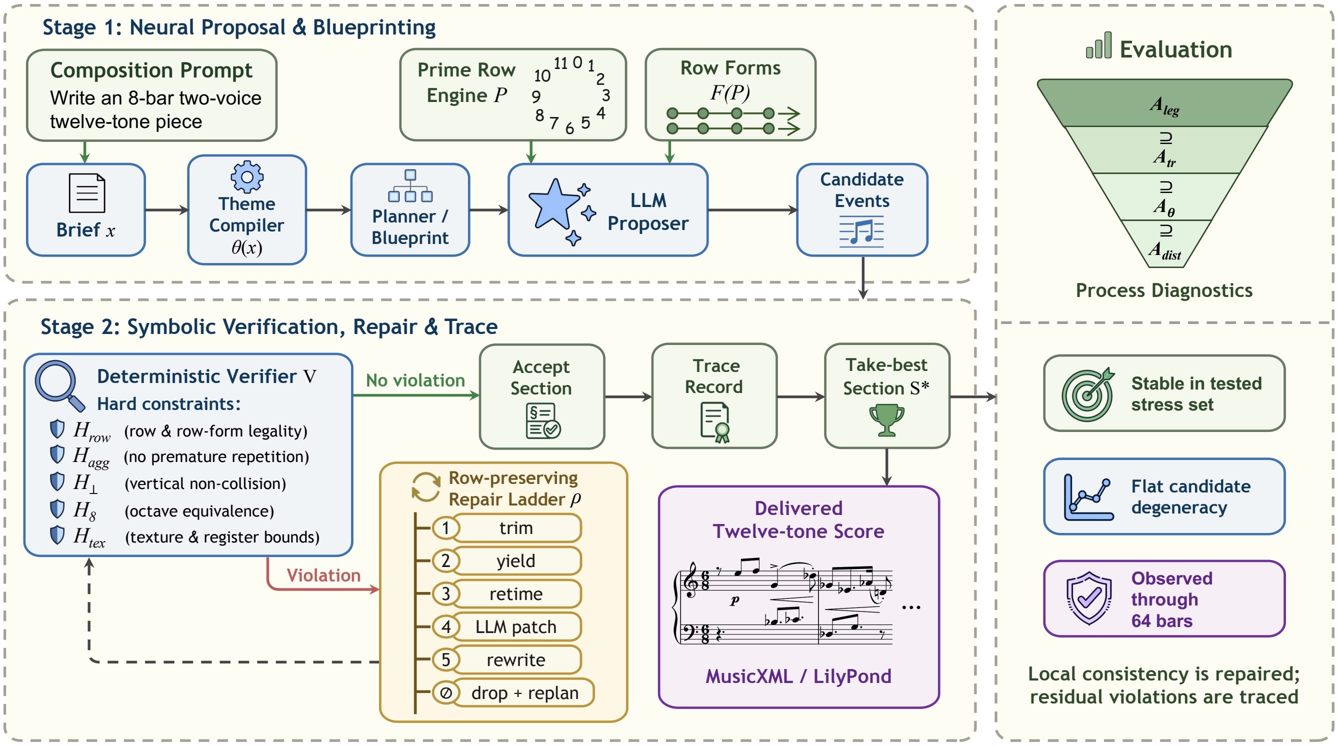

Figure 1: Overview of the harness: an LLM proposer generates candidate events (Stage 1), and a deterministic verifier with a row-aware repair ladder checks or modifies them while recording a per-note trace (Stage 2). A final retained-artefact constraint check gates release (Section 3.5).

tion 3.4); and (3) evaluate, over the fixed task bank, robustness to degeneracy-inducing prompts, length and texture scaling, model capability, cost, and expert preference.

2 Related Work

Symbolic music generation. Prior systems address long-range structure (Huang et al. 2019), controlled infilling (Thickstun et al. 2024), text-to-music attributes (Lu et al. 2023), and multi-track adversarial generation (Dong et al. 2018; Muhamed et al. 2021). They are not organised around a post-hoc verifier that reports domain-level rule violations. SerialGen instead generates twelve-tone rows algorithmically (Feitosa, Santos, and Cardoso 2022), without an LLM proposer or verify–repair loop.

Constrained and grammar-constrained decoding. Prefix-constrained methods reject invalid next tokens using incremental parsers or completion engines (Scholak, Schucher, and Bahdanau 2021; Poesia et al. 2022; Geng et al. 2023). Other approaches use inference-time energies (Qin et al. 2022), training-time logical losses (Ahmed, Chang, and Van den Broeck 2023), or constraint programming followed by LM ranking (Bonlarron and Régin 2024). Our cross-voice, complete-row constraints instead use an external stateful verifier paired with repair.

Verifier-guided generation and self-correction. Related approaches include process reward models (Lightman et al. 2024), generative verifiers (Zhang et al. 2025), self-refinement (Madaan et al. 2023), and feedback-based reflection (Shinn et al. 2023). Unlike these model-mediated approaches, our harness pairs deterministic symbolic checking with explicit repair; Soft rules remain a one-shot, no-verifier baseline.

Reward hacking and specification gaming. Optimising an imperfect reward proxy can reduce performance under the true objective (Amodei et al. 2016; Skalse et al. 2022; Gao, Schulman, and Hilton 2023). Examples include capability-threshold failures (Pan, Bhatia, and Steinhardt 2022) and reward-model over-optimisation (Gao, Schulman, and Hilton 2023); narrower rewards are not generally unhackable (Skalse et al. 2022). We test analogous prompted shortcuts using delivered-score metrics rather than process diagnostics alone.

3 Problem Formulation and Methodology

3.1 Design Rationale

Twelve-tone music as a target. Twelve-tone composition suits generate–verify–repair because its serial operations and aggregate structure are precisely defined (Perle 1991; Babbitt 1960). We operationalise them as predicates over timed events, adding explicit row-form tags, aggregate completion, and cross-voice non-collision checks (Section 3.3). A per-voice row cursor lets repair project a candidate onto its tagged row slice; the LLM handles planning, voicing, and rhythm, while the symbolic layer checks hard rules and records per-note traces.

Event representation. Internally, a composition is a multiset of timed events with MIDI-integer pitches (Section 3.3). The compact schema exposes pitch class $m \bmod 12$, onset, duration, voice, row tag, and register directly to the predicates and repair operators, avoiding notation-format parsing.

Notation and rendering. Composition, verification, and repair use only the canonical event record. Our renderer deterministically exports the same retained events to LilyPond and MusicXML. All blind-evaluation conditions share this implementation path, while corpus-distance analysis uses the same monophonic pitch-line reduction (Appendix A).

3.2 Pitch Classes and the Row-Form Group

Let $\mathbb{Z}_{12} = \{0, 1, \ldots, 11\}$ be the set of pitch classes under octave equivalence (Straus 2016). A *prime row* (tone row) is an ordered aggregate $P = (P_0, \ldots, P_{11})$, i.e. a bijection $\{0, \ldots, 11\} \to \mathbb{Z}_{12}$; every pitch class occurs exactly once (Schoenberg 1950; Perle 1991). Three canonical operators act on rows:

$$\text{transposition} \quad T_n(x) = (x + n) \bmod 12, \tag{1}$$

$$\text{inversion} \quad I(x) = (-x) \bmod 12, \tag{2}$$

$$\text{retrograde} \quad R(x_0, \ldots, x_{11}) = (x_{11}, \ldots, x_0). \tag{3}$$

T_n and I act pointwise on a row; R reverses its time order. Writing $\tilde{P} = T_{-P_0}(P)$ for the *zeroed* prime (so $\tilde{P}_0 = 0$), the four base forms, P (prime), I (inversion), R (retrograde), and RI (retrograde inversion), are

$$b_P = \tilde{P}, \;\; b_I = I \circ \tilde{P}, \;\; b_R = \mathrm{rev}(\tilde{P}), \;\; b_{RI} = \mathrm{rev}(I \circ \tilde{P}), \tag{4}$$

and the *row forms* are their transpositions

$$\mathcal{F}(P) = \{f_{X,n} := T_n \circ b_X : X \in \{P, I, R, RI\},\, n \in \mathbb{Z}_{12}\}. \tag{5}$$

These are the standard four row-form families (Babbitt 1960; Perle 1991). Since $\langle T_n, I \rangle$ is the dihedral group D_{12} of order 24 and R commutes with both pitch-class operations, retrograde contributes an independent factor of 2. The transformation operations therefore form $D_{12} \times C_2$, of order 48. Their action on a row produces at most 48 distinct row forms, because symmetries may cause forms to coincide; hence $|\mathcal{F}(P)| \leq 48$. Each $f_{X,n} = (f_{X,n}[0], \ldots, f_{X,n}[11])$ is itself an aggregate.

Two structural properties of P are used by setup-time constraints. Following the standard notion of derived rows (Babbitt 1960), we call P a *derived row* when its disjoint cells are related by standard serial transformations. For reproducible evaluation, our implementation operationalises this notion for cell sizes $c \in \{3, 4, 6\}$ ($c \mid 12$) by requiring every disjoint length-c block in the partition of P to share the canonical normalisation obtained by taking the minimum over {forward, retrograde, inverted, inverted-retrograde}. Two forms f, g are *hexachordally combinatorial* if their first hexachords partition the aggregate: $\{f[0..5]\} \cap \{g[0..5]\} = \varnothing$ and $\{f[0..5]\} \cup \{g[0..5]\} = \mathbb{Z}_{12}$ (Babbitt 1960).

3.3 Compositions, Traces, and Core Event Consistency

An *event* is a tuple $e = (\tau, \delta, \vec{m}, v, \phi, \sigma)$ with onset $\tau \in \mathbb{Q}_{\geq 0}$, duration $\delta > 0$, MIDI pitches $\vec{m} \in \mathbb{Z}^k$ ($k \geq 1$; $k > 1$ is a chord), a voice label v, a row-form tag $\phi \in \mathcal{F}(P)$, and an ordered tuple of row indices $\sigma = (s_1, \ldots, s_r)$. Its *pitch-class content* is $\mathrm{pc}(e) = \{m \bmod 12 : m \in \vec{m}\}$ and its *sounding interval* is $\iota(e) = [\tau, \tau + \delta)$. A *composition* is a finite multiset of events $S = \{e_1, \ldots, e_N\}$; its restriction to voice v, written S_v, is the event stream of that voice ordered by onset. For a retained candidate, $\widetilde{\mathbf{p}}_{\mathcal{T}}(e)$ denotes the ordered pitch-class tuple stored separately in the trace; unlike $\mathrm{pc}(e)$, it is not defined by reducing $\vec{m}$ modulo 12.

Hard constraints. For fixed prime row P and blueprint B, each hard rule is a predicate $H_{P,B} : S \mapsto \{\top, \bot\}$. The formal target set $\mathcal{H}$ is summarised in Table 5 (Appendix B); the online harness records attempted enforcement, while the final release check evaluates a broader subset directly on retained artefacts. The principal target predicates are:

- **Tagged row consistency** (H_{row}): each retained event satisfies $\mathrm{pc}(e) = \{f_\phi[s] : s \in \sigma\}$ for its tagged (ϕ, σ), and the ordered tags are monotone in the scheduled row-form stream. Because an unrepaired proposal may be dropped while its cursor advances, the retained line can be a tag-preserving subsequence of that stream; it need not be a complete concatenation of forms.
- **Aggregate / no premature repetition** (H_{agg}): within every uninterrupted retained row cycle, no pitch class recurs before the cycle advances, except when the online state admits a local (oscillation/tremolo) window. In the implementation this window is inferred from recent cursor history rather than pre-authorised by a separate blueprint field. Aggregate *completion* is reported separately because dropped events can leave a cycle incomplete.
- **Vertical non-collision** ($H_\perp$): for events e, e' in distinct voices with overlapping sounding intervals ($|\iota(e) \cap \iota(e')| > \epsilon$), $\mathrm{pc}(e) \cap \mathrm{pc}(e') = \varnothing$; the positive tolerance ϵ guards against floating-point boundary contact. The process verifier uses $\epsilon = 10^{-6}$ beats; the independent comparison checker uses the more conservative 10^{-3}-beat threshold reported with that metric.
- **Octave equivalence** (H_8): the independently stored trace tuple agrees with the retained MIDI pitches,

$$H_8(e, \mathcal{T}) \iff \widetilde{\mathbf{p}}_{\mathcal{T}}(e) = (m_1 \bmod 12, \ldots, m_k \bmod 12).$$

- **No global rest / voice independence** (H_{tex}): in sections with at least two active voices, at every time in the active span at least one voice sounds; and the fraction of noteheads participating in any simultaneous higher/lower-voice register inversion is at most 0.1.

Setup-time predicates (H_{seg}, H_{der}, H_{comb}) check segmentation length consistency, derived-row identity, and hexachordal combinatoriality against P and the blueprint.

Algorithm 1 Generate–verify–repair candidate construction for one section: release occurs only after the final constraint check.

Input: section spec, voice states $\{u_v\}$, row engine on P
Parameter: max attempts K ($K = 3$ with replan; $K = 1$ without)
Output: retained candidate events (release follows the final check)

$S^\star \leftarrow \varnothing;\ m^\star \leftarrow \infty$
$\{u_v^0\} \leftarrow \textsc{Snapshot}(\{u_v\})$
for $\mathrm{att} = 1$ **to** K **do**
  $\{u_v\} \leftarrow \textsc{Clone}(\{u_v^0\})$ {no cursor state leaks across attempts}
  propose candidate events per voice (LLM or deterministic), concurrently
  sort candidates by (τ, v); assign event ids {determinism}
  for each candidate e in order **do**
    $e' \leftarrow \rho^\star(e \mid \mathrm{accepted}, u_v)$ {bounded escalation}
    if $e' \neq \perp$ **then**
      accept e'
    end if
    $\textsc{Advance}(u_v)$ {cursor advances even if dropped}
  end for
  evaluate section-level and multi-voice hard rules; let S_{att} be the result
  if $|\widehat{\mathcal{V}}_{\mathrm{proc}}(S_{\mathrm{att}})| < m^\star$ **then**
    $S^\star \leftarrow S_{\mathrm{att}};\ m^\star \leftarrow |\widehat{\mathcal{V}}_{\mathrm{proc}}(S_{\mathrm{att}})|$
  end if
  if $m^\star = 0$ **then**
    break
  end if
  Replan: reassign row forms to minimise simulated collisions (guarded)
end for
return candidate $S^\star$

Definition 1 (Core event-consistency language) *The* core event-consistency language *for P and B is* $\mathcal{L}(P, B) = \{S : \forall H_{P,B} \in \mathcal{H},\ H_{P,B}(S) = \top\}$.

This target language does not by itself require nonempty output, realisation of every requested voice, or completion of every aggregate; those coverage properties are reported separately. We therefore do not use membership in $\mathcal{L}(P, B)$ as a synonym for complete twelve-tone legality.

The implementation reconstructs the per-voice states from the events accepted before this section at the start of every attempt. Cursor advances made by a failed attempt therefore do not affect later attempts; only the operative section specification may change through replanning. The algorithmic replan may alter the operative section and row-form schedule within an attempt. The retained result currently stores the initial blueprint rather than a fully materialised history of those local replans. Consequently, agreement with the stored schedule is reported separately as a provenance check and is not implied by a process pass.

An ideal final-score verifier $\mathcal{V}$ maps a composition to the multiset of hard violations, $\mathcal{V}(S) = \{(H, w) : H \in \mathcal{H},\ w \text{ witnesses } \neg H(S)\}$, so $S \in \mathcal{L}(P, B) \iff \mathcal{V}(S) = \varnothing$. Soft (aesthetic) rules are collected separately as warnings and never affect membership in $\mathcal{L}$. The implemented process diagnostic instead reads violations accumulated during online generation; write this list as $\widehat{\mathcal{V}}_{\mathrm{proc}}$. It is not a fresh evaluation of every predicate on the final normalised candidate.

Traces. Every accepted event e carries a *certificate* $c_{\mathcal{T}}(e) = (\tau, \delta, \vec{m}, \tilde{\mathbf{p}}_{\mathcal{T}}, v, q, \phi, \sigma, \mathrm{status}, \mathrm{owner})$, where q is the section identifier. Thus MIDI pitches and claimed pitch classes are separately serialised rather than identified by definition. The implementation additionally stores event ids, repair history, and acceptance flags. The trace $\mathcal{T}(S) = \{c_{\mathcal{T}}(e) : e \in S\}$ is a per-event provenance log; it is used by higher process-acceptance tiers (Section 3.5) to test whether a claimed pitch class is derivable from a recorded row position. It supports event-level inspectability, not complete planning provenance or an independent certificate of the entire retained candidate.

3.4 The Generate–Verify–Repair Harness

The harness maps a natural-language brief x to either a candidate passing the specified release check or an explicit failure status. It first constructs a candidate $S^\star$, with the design goal of reducing $|\widehat{\mathcal{V}}_{\mathrm{proc}}(S^\star)|$ to zero within a bounded attempt budget. A theme compiler turns x into a structured specification $\theta(x)$ (bar counts, voice counts, textures, derived / combinatorial requests, etc.), which a planner expands into a blueprint of sections; the prime row P is generated deterministically from the compiled specification once per task and is then held fixed across models, conditions, and the three runs. Within each section the harness runs Algorithm 1; after assembly, the final constraint check of Section 3.5 gates release.

Composer-facing control. The harness also accepts hand-authored blueprints through the same verify–repair procedure. Composers can specify P, up to four voices, and per-voice row forms and schedules, register $[\ell_v, h_v]$, rhythmic density, trichordal/tetrachordal/hexachordal segmentation, optional hexachordal combinatoriality, and staggered entries. These are executable inputs: repair enforces registers, although local replanning may revise schedules. This supports direct control over an individual serial practice, but is not evaluated; the benchmark uses only blueprints automatically synthesised from briefs.

Row-preserving repair. The repair operator is an ordered escalation ladder $\rho = (\rho_1, \ldots, \rho_5)$ applied to a flagged event; $\rho^\star(e)$ returns the first $\rho_j(e)$ that clears the local violation, or $\perp$ (drop the event and defer its slot to a section replan) if none do. The first three rungs target vertical collisions by editing *only timing*:

$$\rho_j(e) = (\tau', \delta', \vec{m}, v, \phi, \sigma), \quad j \in \{1, 2, 3\}, \tag{6}$$

i.e. $\rho_{1\text{–}3}$ change (τ, δ) but leave $(\vec{m}, \phi, \sigma)$ fixed: rung 1 trims the current note's tail, rung 2 yields by trimming an overlapping sustained note in another voice, rung 3 retimes the note inside its own free window. The last two rungs are LLM patches checked again after register clamping (rung 4 edits the single event and rung 5 rewrites its row segment), after which the event is dropped and the section is flagged for replan. The following invariance applies only to the deterministic timing rungs:

Lemma 1 (Repair invariance) *For $j \in \{1,2,3\}$, $\mathrm{pc}(\rho_j(e)) = \mathrm{pc}(e)$ and the tagged (ϕ,σ) are unchanged. Hence the deterministic rungs preserve H_{row} and H_{agg} and do not alter MIDI pitch content. H_8 is evaluated after the repaired event is serialised into the trace. Since the rungs only* shorten *or* move into free space*, they introduce no new vertical collision or intra-voice overlap. Rung 2 additionally* shortens *the overlapping sustained note in the* other *voice; being again only a truncation, it creates no new collision and leaves that voice's pitch classes and row tags intact, so the invariance extends to every event the rung edits.*

Lockstep invariant. Generation and verification maintain a shared row cursor per voice. Even when $\rho^\star(e) = \perp$, the ADVANCE step of Algorithm 1 still increments the cursor, so the verifier's expected row index never lags the generator's.

Proposition 1 (Cursor lockstep) *At every step, for each voice v the verifier-side next index equals the generator-side next index. Consequently a dropped event costs exactly one rest at its slot and does not desynchronise later tags. More generally, a dropped multi-index candidate omits its tagged row slice. The retained stream is therefore a tag-preserving subsequence, not necessarily a complete row statement.*

Bounded best-attempt selection. Let $S^{(1)},\dots,S^{(K)}$ be the per-attempt results and $S^\star = \arg\min_k |\widehat{\mathcal{V}}_{\mathrm{proc}}(S^{(k)})|$ the retained candidate ("take-best").

Proposition 2 (Projection and selection) *(i) Within a fixed attempt and relative to its operative local section state, candidate construction, tag checking, and the deterministic repair rungs preserve each retained event's expected row pitch class and octave-equivalent tag; later events remain aligned after a drop by Proposition 1. This is an event-local invariant and does not imply that every requested aggregate is complete or that the event schedule matches the initial blueprint retained in the artefact. (ii) If $M_K = \min_{1\le k\le K}|\widehat{\mathcal{V}}_{\mathrm{proc}}(S^{(k)})|$, then*

$$M_{K+1} \le M_K, \qquad |\widehat{\mathcal{V}}_{\mathrm{proc}}(S^\star)| = M_K,$$

so retaining an additional attempt cannot worsen the best observed violation count. (iii) No existence claim follows from the bounded search: the retained candidate can contain violations and therefore fail the final release gate.

3.5 Process Diagnostics and Final Release Check

For harness runs, we retain a chain of increasingly strict *process* acceptance predicates over a composition S for brief x, supporting transparent stage-wise diagnostic reporting:

$$A_{\mathrm{leg}}(S) = \big[\widehat{\mathcal{V}}_{\mathrm{proc}}(S) = \varnothing\big], \tag{7}$$

$$A_{\mathrm{tr}}(S) = A_{\mathrm{leg}}(S) \wedge \big[|\mathcal{T}(S)| = |S|\big], \tag{8}$$

$$A_{\theta}(S) = A_{\mathrm{tr}}(S) \wedge \Theta\big(S, \theta(x)\big), \tag{9}$$

$$A_{\mathrm{dist}}(S) = A_{\theta}(S) \wedge C_{\mathrm{dist}}(S), \tag{10}$$

Here Θ checks bar/voice counts and requested techniques, while C_{dist} requires attacks in at least 75% of each active voice's bars. The diagnostics are nested:

$$A_{\mathrm{dist}} \Rightarrow A_{\theta} \Rightarrow A_{\mathrm{tr}} \Rightarrow A_{\mathrm{leg}}. \tag{11}$$

These tiers reuse process records and are diagnostics, not release decisions. After score assembly, the separately implemented final constraint check A_{check} (Appendix B) rescans the retained candidate. The output contract is

$$\textsc{Compose}(x) = \begin{cases} S^\star, & A_{\mathrm{check}}(S^\star) = 1, \\ \textsc{Failure}, & \text{otherwise.} \end{cases}$$

The gate does not trigger further generation; failed candidates remain stored only for analysis. Stored-blueprint agreement is reported separately because operative replans are not fully materialised.

4 Experimental Design

Tasks. We use 40 single-factor tasks: a 20-task length sweep (8–64 bars, two voices) and a 20-task texture sweep (1–4 voices, eight bars), sharing a two-voice/eight-bar anchor. Each task contains one mild writing concern. Full briefs and inventories appear in Appendix C.

Models. DeepSeek V4 Pro/Flash, GPT-5 Nano, and Qwen3 235B-A22B (Qwen3 235B) run both raw and in-harness. The full model inventory appears in Appendix C.

Expert evaluation. Five experts compare 24 anonymised Qwen3 235B pairs (8 tasks $\times$ three contrasts).

Experimental setup. The four paired models share one API and decoding configuration (Appendix D). A task's prime row is fixed across models, conditions, and runs; seeds control only LLM sampling. After averaging runs within task, we report point estimates and 2000-resample nonparametric family-clustered bootstrap (Efron 1979) confidence intervals (CIs). Each draw samples the ten prompt families with their four tasks. The ablation study comprises the deterministic backbone, raw generation, Self-Refine, Soft rules, and the full harness.

Metrics. We report process diagnostics, constraint-wise marginal pass rates, constraint-checked delivery yield, and a narrower, separately implemented collision and serialisation-consistency check. Other outcomes are corpus squared maximum mean discrepancy (MMD^2) (Gretton et al. 2012) with a Gaussian radial basis function (RBF) kernel and Kullback–Leibler (KL) divergence (Kullback and Leibler 1951) (Appendix A), degeneracy (Appendix B), cost (Appendix H), and expert preference (Appendix I). Red-team analysis combines degeneracy, retained material, and the independent check rather than relying on process fields.

5 Results and Analysis

5.1 Length and Texture Scaling

Figure 2 separates model and seed variation in the local collision diagnostic. Across plotted settings, harness three-run means remain at or below 2.05%, whereas raw three-run means, conditional on parseability, reach 14.76%. Raw

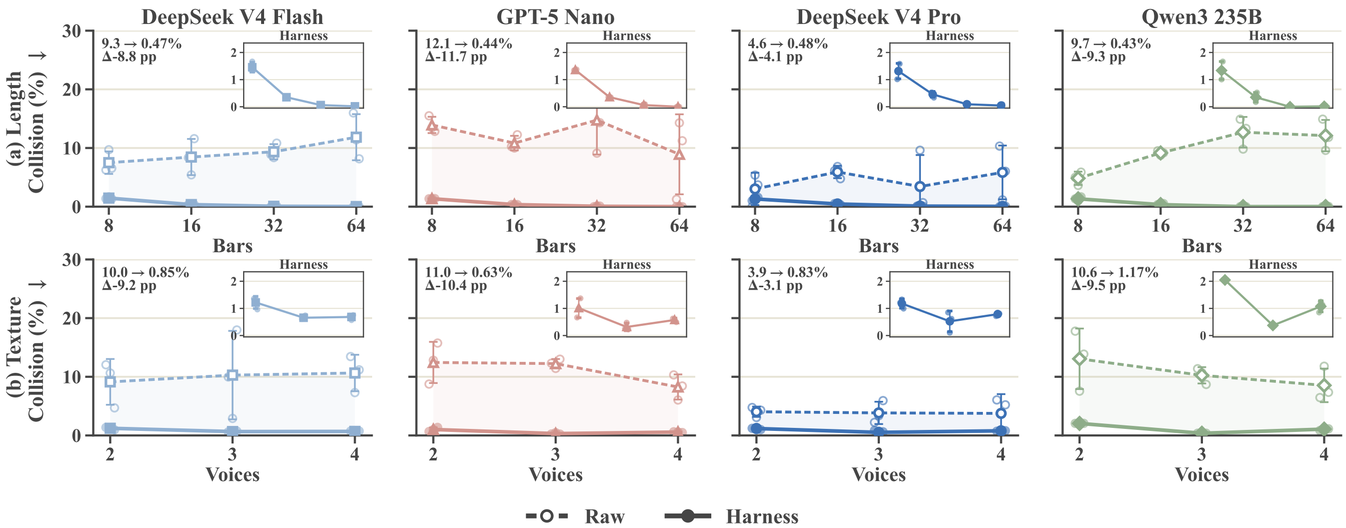

Figure 2: Cross-voice collision rates by model. Rows vary length (top) and texture (bottom); columns separate models. Curves show three-run means ± one standard deviation, faint points show runs, and insets magnify harness rates. Raw rates are conditional on parseability; one voice is omitted because no cross-voice denominator exists.

Table 1: Ablation and prompt-only deltas versus matching full-harness baselines. The top block is deterministic; lower blocks use Qwen3 235B. Values cover 40 paired tasks; "accept" is a process outcome, not an independent legality measure.

Condition	Δ process accept	Δ ind. core	Δ deg.
Deterministic backbone (vs. full harness)			
No repair	−0.192	+0.050	−0.006
No repair + no replan	−0.192	+0.025	−0.004
No replan	−0.025	+0.025	+0.001
LLM in the loop (vs. full harness)			
No skills	+0.017	−0.042	+0.000
Fixed row	−0.008	−0.092	+0.001
Prompt-only baselines (vs. full harness)			
Self-refine	−0.183	−0.333	+0.241
Soft rules	−0.237	−0.400	+0.208

length trends are model-dependent; harness rates generally fall toward zero as length grows. Rates are opportunity-normalised (Appendix B). Figure 3 complements this local metric with an end-to-end filtering sequence over all 480 attempted tasks. From raw to harness, parseable output (*Valid*) rises from 410/480 (85.4%) to 480/480; the independent collision and serialisation check (*Core*) rises from 33.5% to 58.3%; and constraint-checked delivery (*Full*) rises from 64/480 (13.3%) to 231/480 (48.1%). Every model improves at Core and Full. The harness abstains on the remaining 249 candidates rather than releasing failed scores. Full rates and denominator caveats appear in Appendix B; red-team parseability appears in Appendix F. This primary release check is schedule-relaxed: 184/480 candidates disagree with the initially stored schedule, including 61 that pass every other check. Requiring stored-blueprint agreement reduces delivery to 170/480 (35.4%).

Constraint-wise diagnosis. Table 1 shows that repair raises process acceptance but not the narrow candidate check. Without repair, candidates contain 11.2 fewer events (0.71 fewer notes/bar) on average, consistent with reduced collision exposure; coverage changes by only −0.001. Thus, the +0.050 narrow-check delta does not establish that repair harms candidate consistency. More details appear in Appendix G. At the full-system level, marginal rates show that the harness passes row-form/order (99.4%), no premature repetition (99.8%), aggregate completion (95.6%), and the voice-crossing bound (96.0%) in nearly all candidates. The remaining bottlenecks are vertical non-collision (58.3%) and the combined texture check (74.6%). Vertical non-collision determines the equal-valued narrow collision and serialisation-consistency conjunction because its other components pass for every harness candidate. Thus, the 48.1% delivery yield is a conjunction across heterogeneous checks, not typical per-constraint performance (Appendix B, Table 6).

5.2 Robustness to Degeneracy-Inducing Prompts

Tables 10 and 11 (Appendix F) report the exploratory end-to-end prompt stress test on Qwen3 235B. Because attack suffixes also pass through the theme compiler, they can change the executable specification rather than attacking a fixed proposer alone. Harness degeneracy stays near 0.05 and its collision and serialisation-consistency pass rate remains at 0.55–0.65, but the *Long chords* variant eliminates theme coverage.

5.3 Capability–Scaffolding Interaction and Cost

Across four models, in-harness generation raises the independent collision and serialisation pass rate to 0.53–0.63, constraint-checked delivery to 0.43–0.52, and lowers degeneracy to 0.047–0.053 (Table 17). These descriptive gains

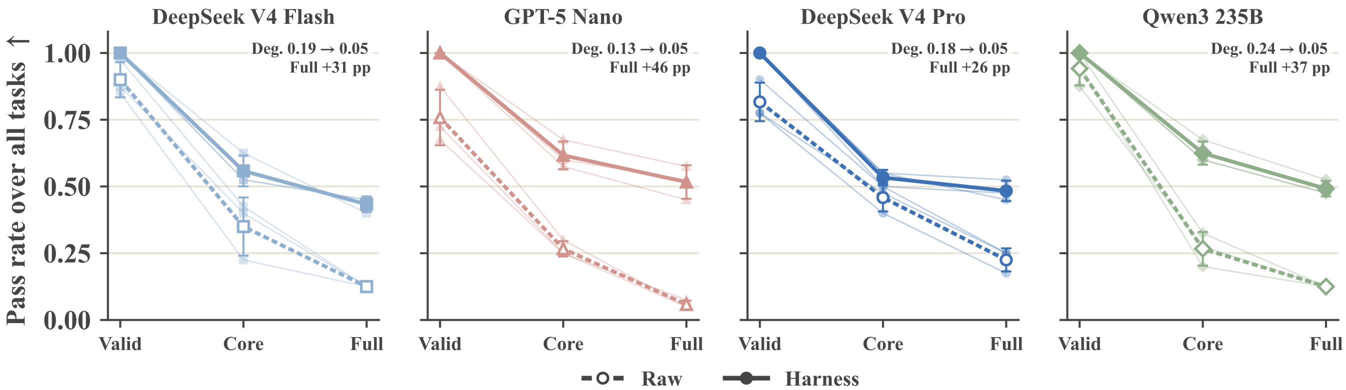

Figure 3: Raw-to-harness reliability ladder by model: parseable output (*Valid*), independent collision and serialisation pass (*Core*), and the schedule-relaxed release check (*Full*). Faint lines show runs; bold lines show means ± one standard deviation. Annotations give degeneracy (Deg.) and the all-task Full-rate gain.

Table 2: Mean per-run cost on Qwen3 235B ($n \approx 110$–120). Tokens are provider-reported total usage; wall-clock covers end-to-end execution.

Condition	LLM calls	Tokens	Wall-clock (s)
Blueprint only	1.0	2,433	24.0
Raw LLM	1.0	5,257	76.7
Soft rules	1.0	5,117	62.8
Self-refine	5.0	23,850	223.9
Full harness, no skills	627.3	234,845	814.5
Full harness	638.0	823,482	841.0

apply only to the tested models and are costly: on Qwen3 235B, the harness uses 638 calls, 823k tokens, and 841 seconds per run, versus one call, 5.3k tokens, and 77 seconds for raw generation (Table 2).

5.4 Musical Quality

Conditional on parseability, mean degeneracy falls for every model from 0.132–0.244 raw to 0.047–0.053 in-harness (Figure 3; Table 17). Because this diagnostic captures structural collapse rather than musical quality, Figure 4 provides a representative score.

Corpus distribution analysis. Table 3 compares the retained harness candidates with a 20-piece human corpus using pitch-distribution features (Appendix A). Harness candidates are closer to the strict-serial than the free-atonal subset under both MMD^2 and KL.

Task-completion analysis. Table 4 captures properties omitted by the corpus features. Every harness candidate is parseable, covers 97.3% of requested voice–bar cells, and completes aggregates in 95.6% of all runs; parseable raw candidates cover only 63.5% of voice–bar cells.

5.5 Expert Blind Evaluation

Five experts evaluate 24 anonymised retained-candidate pairs. The full harness receives 82–90% of choices over raw generation for adherence, perceived legality, coherence, and overall quality, and 70% for style. Against the ablations, full-harness win rates span 55–75% without skills and 57–75% without repair. Task-clustered intervals and low inter-rater agreement qualify these aggregate preferences (Appendix I).

Table 3: Corpus distribution distance for 480 retained harness candidates. One RBF bandwidth is shared. KL is interval-class / directed-interval / pitch-class, generated || corpus.

Corpus subset	n	MMD^2	KL (ic / di / pc)
all	20	0.330	0.073 / 0.074 / 0.009
strict-serial	9	0.278	0.041 / 0.042 / 0.004
free-atonal	11	0.419	0.145 / 0.155 / 0.024

Table 4: Raw vs. harness candidate completion over 480 paired cells. Values are three-run mean ± standard deviation pooled over four models. Parseability and aggregate completion use the intention-to-treat denominator; coverage and voice realisation are conditional on parseable candidates.

Metric	Raw	Harness
Parseable-candidate rate	0.854 ± 0.055	1.000 ± 0.000
Active-bar coverage	0.635 ± 0.020	0.973 ± 0.001
Voice realisation	0.995 ± 0.005	1.000 ± 0.000
Aggregate-completion pass	0.673 ± 0.046	0.956 ± 0.006

6 Conclusion and Limitations

We introduce a generate–verify–repair harness for constrained twelve-tone composition. It raises schedule-relaxed, constraint-checked delivery from 13.3% to 48.1% across four paired models, but abstains on 51.9% of runs; on Qwen3 235B, it averages 638 calls and 823k tokens per run. Its trace supports event-level inspectability, not complete planning provenance, and stored-blueprint agreement lowers delivery to 35.4%. The contribution is therefore improved selective delivery and explicit failure within the tested task bank, not production-ready or reliably complete composition.

Acknowledgements

This work is supported by the Advanced Discipline Construction Project of Beijing Universities, the Special Programme of National Natural Science Foundation of China (Grant No. T2341003), and the Major Programme of National Social Science Fund of China (Grant No. 21ZD19).

References

Ahmed, K.; Chang, K.-W.; and Van den Broeck, G. 2023. A Pseudo-Semantic Loss for Autoregressive Models with Logical Constraints. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Amodei, D.; Olah, C.; Steinhardt, J.; Christiano, P.; Schulman, J.; and Mané, D. 2016. Concrete Problems in AI Safety. arXiv:1606.06565.

Babbitt, M. 1960. Twelve-Tone Invariants as Compositional Determinants. *The Musical Quarterly*, 46(2): 246–259.

Bonlarron, A.; and Régin, J.-C. 2024. Intertwining CP and NLP: The Generation of Unreasonably Constrained Sentences. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 7600–7608.

Dong, H.-W.; Hsiao, W.-Y.; Yang, L.-C.; and Yang, Y.-H. 2018. MuseGAN: Multi-track Sequential Generative Adversarial Networks for Symbolic Music Generation and Accompaniment. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, volume 32, 34–41.

Efron, B. 1979. Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics*, 7(1): 1–26.

Feitosa, M.; Santos, R. S.; and Cardoso, L. S. 2022. An Intervallic Approach to Tone Rows: SerialGen, a Twelve-Tone Row Generator. In *Proceedings of the 6th International MusMat Conference*, 132–143. Rio de Janeiro, Brazil: Universidade Federal do Rio de Janeiro.

Gao, L.; Schulman, J.; and Hilton, J. 2023. Scaling Laws for Reward Model Overoptimization. In *International Conference on Machine Learning (ICML)*, 10835–10866.

Garreau, D.; Jitkrittum, W.; and Kanagawa, M. 2017. Large Sample Analysis of the Median Heuristic. arXiv:1707.07269.

Geng, S.; Josifoski, M.; Peyrard, M.; and West, R. 2023. Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 10932–10952.

Gretton, A.; Borgwardt, K. M.; Rasch, M. J.; Schölkopf, B.; and Smola, A. 2012. A Kernel Two-Sample Test. *Journal of Machine Learning Research*, 13(25): 723–773.

Huang, C.-Z. A.; Vaswani, A.; Uszkoreit, J.; Shazeer, N.; Simon, I.; Hawthorne, C.; Dai, A. M.; Hoffman, M. D.; Dinculescu, M.; and Eck, D. 2019. Music Transformer: Generating Music with Long-Term Structure. In *International Conference on Learning Representations (ICLR)*.

Kullback, S.; and Leibler, R. A. 1951. On Information and Sufficiency. *The Annals of Mathematical Statistics*, 22(1): 79–86.

Lightman, H.; Kosaraju, V.; Burda, Y.; Edwards, H.; Baker, B.; Lee, T.; Leike, J.; Schulman, J.; Sutskever, I.; and Cobbe, K. 2024. Let's Verify Step by Step. In *International Conference on Learning Representations (ICLR)*.

Lu, P.; Xu, X.; Kang, C.; Yu, B.; Xing, C.; Tan, X.; and Bian, J. 2023. MuseCoco: Generating Symbolic Music from Text. arXiv:2306.00110.

Madaan, A.; Tandon, N.; Gupta, P.; Hallinan, S.; Gao, L.; Wiegreffe, S.; Alon, U.; Dziri, N.; Prabhumoye, S.; Yang, Y.; Gupta, S.; Majumder, B. P.; Hermann, K.; Welleck, S.; Yazdanbakhsh, A.; and Clark, P. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Muhamed, A.; Li, L.; Shi, X.; Yaddanapudi, S.; Chi, W.; Jackson, D.; Suresh, R.; Lipton, Z. C.; and Smola, A. J. 2021. Symbolic Music Generation with Transformer-GANs. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, volume 35, 408–417.

Pan, A.; Bhatia, K.; and Steinhardt, J. 2022. The Effects of Reward Misspecification: Mapping and Mitigating Misaligned Models. In *International Conference on Learning Representations (ICLR)*.

Perle, G. 1991. *Serial Composition and Atonality: An Introduction to the Music of Schoenberg, Berg, and Webern*. Berkeley, CA: University of California Press, 6th edition.

Poesia, G.; Polozov, O.; Le, V.; Tiwari, A.; Soares, G.; Meek, C.; and Gulwani, S. 2022. Synchromesh: Reliable Code Generation from Pre-trained Language Models. In *International Conference on Learning Representations (ICLR)*.

Qin, L.; Welleck, S.; Khashabi, D.; and Choi, Y. 2022. COLD Decoding: Energy-based Constrained Text Generation with Langevin Dynamics. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Schoenberg, A. 1950. Composition with Twelve Tones. In Newlin, D., ed., *Style and Idea*, 102–143. New York: Philosophical Library.

Scholak, T.; Schucher, N.; and Bahdanau, D. 2021. PICARD: Parsing Incrementally for Constrained Auto-Regressive Decoding from Language Models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 9895–9901.

Shinn, N.; Cassano, F.; Gopinath, A.; Narasimhan, K.; and Yao, S. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Skalse, J.; Howe, N. H. R.; Krasheninnikov, D.; and Krueger, D. 2022. Defining and Characterizing Reward Hacking. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Straus, J. N. 2016. *Introduction to Post-Tonal Theory*. New York: W. W. Norton & Company, 4th edition.

Thickstun, J.; Hall, D. L. W.; Donahue, C.; and Liang, P. 2024. Anticipatory Music Transformer. *Transactions on Machine Learning Research*.

Zhang, L.; Hosseini, A.; Bansal, H.; Kazemi, M.; Kumar, A.; and Agarwal, R. 2025. Generative Verifiers: Reward Modeling as Next-Token Prediction. In *International Conference on Learning Representations (ICLR)*.

A Corpus-Distance Features and Metrics

This appendix gives the full feature schema and distance definitions summarised in Section 5.4. Everything is computed from monophonic pitch lines so that generated candidates and the human corpus are compared on identical footing.

Pitch lines. Human scores are stored as MusicXML and parsed once with `music21`. Each staff/part is split by explicit MusicXML voice; without voice elements, it forms one line. Notes are ordered by measure and within-measure offset, rests are omitted, chords retain only their highest MIDI pitch, and lines shorter than two notes are discarded, yielding monophonic lines $\mathcal{P} = \{\ell\}$. Generated records undergo the same onset ordering, chord reduction, and length filter.

Feature vector. For a line $\ell = (p_1, \ldots, p_T)$ we accumulate, over all lines of a piece, the pitch classes $p_t \bmod 12$ and the nonzero directed intervals $d_t = (p_{t+1} - p_t) \bmod 12$, with interval class $\mathrm{ic}(d) = \min(d, 12 - d)$. Normalising each histogram to a probability vector yields three distribution blocks: interval class $\mathbf{c} \in \Delta^5$ (dims 1–6), directed interval $\mathbf{d} \in \Delta^{10}$ (dims 1–11), and pitch class $\mathbf{h} \in \Delta^{11}$ (dims 0–11). Four melodic scalars are appended: the normalised Shannon entropy of $\mathbf{d}$, the fraction of the 11 directed intervals that occur, the *leap ratio* (fraction of absolute steps $|p_{t+1} - p_t| \geq 8$), and the mean absolute interval divided by 12 (capped at 1). Concatenation gives a fixed $\phi(\cdot) \in \mathbb{R}^{33}$ $(6 + 11 + 12 + 4)$.

Squared MMD. Let $X = \{x_i\}_{i=1}^n$ (corpus) and $Y = \{y_j\}_{j=1}^m$ (generated) be the feature vectors. With the RBF kernel $k(a, b) = \exp(-\gamma \|a - b\|^2)$ we report the unbiased estimator of squared MMD (Gretton et al. 2012)

$$\widehat{\mathrm{MMD}}_u^2 = \frac{\sum_{i \neq j} k(x_i, x_j)}{n(n-1)} + \frac{\sum_{i \neq j} k(y_i, y_j)}{m(m-1)} - \frac{2}{nm} \sum_{i,j} k(x_i, y_j),$$

where $\gamma = 1/\tilde{s}$ and $\tilde{s}$ is the median squared distance over distinct unordered pairs in the fixed pool containing the full 20-piece corpus and all 480 harness candidates, using the median-heuristic convention analysed by Garreau, Jitkrittum, and Kanagawa (2017). The resulting $\gamma = 19.639328$ is reused for every corpus subset and model comparison. Smaller estimated MMD^2 generally indicates greater similarity in the feature distribution; negative values are sampling fluctuations around zero, not negative distances. For a characteristic kernel, the population MMD is zero if and only if the two feature distributions are identical (Gretton et al. 2012).

Per-block KL. For each distribution block $B \in \{\mathbf{c}, \mathbf{d}, \mathbf{h}\}$ we use the directed KL divergence (Kullback and Leibler 1951): we average the per-piece block vectors within each set to obtain mean distributions $\bar{B}^{\mathrm{gen}}$ and $\bar{B}^{\mathrm{cor}}$, add a smoothing constant $\varepsilon = 10^{-9}$, renormalise, and report $\mathrm{KL}(\bar{B}^{\mathrm{gen}} \,\|\, \bar{B}^{\mathrm{cor}})$. The three values reported as "KL (ic / di / pc)" in Table 3 are exactly these block divergences.

Protocol. The generated set comprises 480 retained harness candidates across the four harness-tier models; the human set is the 20 frozen corpus pieces. Style subsets (strict-serial vs. free-atonal) partition the corpus by piece. Feature extraction, MMD^2, and KL are deterministic post-hoc computations on stored pitch lines.

Observed distances and interpretation. Harness candidates are closer to the strict-serial subset (MMD^2=0.278; KL 0.041/0.042/0.004) than to the free-atonal subset (MMD^2=0.419; KL 0.145/0.155/0.024); per-model MMD^2 ranges from 0.293 to 0.353. These distances have no uncertainty estimate, and the style subsets confound composer, work, and genre, so the comparison cannot isolate serial practice or establish human-level quality. A sample score appears in Figure 4.

B Twelve-Tone Harness Metrics

This appendix defines the scalar metrics summarised in Section 4. Process acceptance tiers (A_{leg}–A_{dist}) are formalised in Section 3.5; here we detail the delivery metrics used alongside them. Table 5 restates the hard constraint predicates $\mathcal{H}$ that define the core event-consistency target (Section 3.3).

Table 5: Hard constraint predicates $\mathcal{H}$ defining the core event-consistency language $\mathcal{L}(P, B)$. Membership does not require nonempty output, requested-voice realisation, or aggregate completion; soft (aesthetic) rules are excluded.

Predicate	Condition
H_{row}	each event's pitch-class set matches its tagged row-form slice $f_\phi[\sigma]$
H_{agg}	within a retained row cycle, no pitch class repeats before the cursor advances (per voice)
$H_{\perp}$	time-overlapping events in distinct voices share no pitch class
H_8	trace pitch-class tuple equals the retained MIDI tuple modulo 12
H_{tex}	in multivoice sections, no global rest; $\leq 10\%$ of noteheads participate in register inversions
H_{seg}	segment slice lengths consistent, no cross-segment spill
H_{der}	derived-row cell identity (when requested)
H_{comb}	hexachordal combinatoriality of paired forms (when requested)

Independent collision and serialisation-consistency pass rate. Let S^{ret} be the multiset of *accepted* events retained in a candidate (events rejected during verify–repair are excluded). The independent collision and serialisation-consistency pass rate is the mean of a binary score computed by a separate checker without calling the harness verifier. It covers the following issues:

- **Vertical non-collision**: distinct voices that overlap in time by more than 10^{-3} beats must not share a pitch class; no event may repeat a pitch class within a chord. The overlap tolerance excludes floating-point boundary dust from tuplet durations while retaining musically meaningful simultaneity.

- **Serialisation consistency**: every retained pitch class equals its MIDI pitch modulo 12, and every event has matching trace pitches, voice, section, row form, and row indices.

For the scaling analysis, let $\mathcal{O}$ contain all unordered retained-event pairs from distinct voices whose sounding intervals overlap by more than 10^{-3} beats. We report the opportunity-normalised collision rate

$$R_{\mathrm{ov}}(S) = 100\,\frac{\sum_{(e_i,e_j)\in\mathcal{O}} \mathbf{1}[\mathrm{pc}(e_i)\cap\mathrm{pc}(e_j)\neq\varnothing]}{|\mathcal{O}|}. \quad (12)$$

Each overlapping event pair contributes one opportunity regardless of overlap duration or chord size. Within-event duplicate pitch classes are excluded from R_{ov}, and the rate is undefined when $\mathcal{O}$ is empty; the one-voice figure point is displayed at zero solely as a plotting convention. For every model, run, and axis setting, we sum the numerator and denominator over the five corresponding task outputs before division, then report the mean and sample standard deviation of the three run-level pooled ratios. Thus, opportunity-rich compositions receive proportionally greater weight. The trace is serialised directly from accepted events, so this component checks record completeness and copy consistency rather than independently validating the musical row claim. Thus, "serialisation consistency" here means agreement among redundant event, MIDI, and trace fields; it is not serial row legality. The substantive retained-note component of the narrow score is therefore collision detection; row-form and schedule consistency are left to the final constraint check below. For the original six-model benchmark, this narrow score is the only event check available uniformly across raw and harness candidates.

Final constraint check and release gate. For retained result artefacts, a second binary check independently reconstructs all 48 row forms and checks event pitches against their tagged indices and within-voice row cycles. It also checks premature repetition; completion of at least one 12-tone aggregate per active voice (while logging a trailing partial cycle separately); segmentation boundaries; derived-row identity; hexachordal combinatoriality; global rests; the 10% voice-crossing bound; and the core checks above. Events carrying implementation-derived oscillation or boundary-elision tags may repeat their local indices. No implementation from the main verifier or row engine is called. The primary final-check result excludes only comparison with the initially stored row-form schedule, because the retained blueprint does not materialise operative replans. A stricter provenance variant includes that comparison. Aggregate completion is stricter than the core event-consistency language $\mathcal{L}(P,B)$ defined above, where incomplete trailing cycles are permitted; accordingly, this result is not presented as the pass rate of $\mathcal{L}(P,B)$.

Applied to the 480 retained harness candidates, the schedule-relaxed final gate releases 231 (48.1%) and returns explicit failure for 249; per-model release rates range from 43.3% to 51.7%. The stricter stored-blueprint-agreement variant passes 170 (35.4%). Schedule disagreement appears in 184 candidates, of which 61 pass every other component. The constraint-wise rates in Table 6 localise the remaining failures; categories overlap. Derived-row and combinatoriality checks have no applicable requested tasks in this benchmark. The gate is deterministic and introduces no additional LLM calls. It does not estimate the success or cost of check-triggered retries, which are not used.

Table 6: Marginal and joint final-check pass rates over 480 paired raw and harness runs. Unparseable raw candidates count as failures. Marginal checks overlap and therefore do not multiply to the joint rate. The hard conjunction excludes aggregate completion and stored-schedule agreement; the release gate adds aggregate completion.

Check component	Raw	Harness
Vertical non-collision	33.5%	58.3%
Row-form/order	27.1%	99.4%
No premature repetition	85.2%	99.8%
Aggregate completion	67.3%	95.6%
Segmentation	85.4%	99.8%
Texture (both checks)	69.0%	74.6%
No global rest	74.4%	77.7%
Voice-crossing bound	79.6%	96.0%
Hard-constraint conjunction	14.2%	49.0%
Schedule-relaxed release gate	13.3%	48.1%
with stored-schedule agreement	7.9%	35.4%

Task-completion diagnostics. Table 4 uses all 480 same-model, task, setting, and seed pairs. Active-bar coverage averages, over requested voices, the fraction of requested bars containing at least one attack; voice realisation is the fraction of requested voices present. These two means are conditional on parseable candidates. Parseability and aggregate-completion pass rates use all runs, counting invalid raw candidates as failures. We pool the four models within each seed and report the mean and sample standard deviation across three seeds.

Scaling details. Across models, collision-bearing overlap rates are 1.01–2.05% for two voices, 0.32–0.66% for three, and 0.57–1.08% for four. Three length-series decrease monotonically; Qwen3 235B moves from zero at 32 bars to 0.015% at 64. The one-voice setting is omitted because it has no defined cross-voice denominator. Rates count overlapping event pairs equally and do not weight overlap duration or chord size; overlap exposure can itself change with generation.

Degeneracy score. The degeneracy score $D(S)\in[0,1]$ aggregates five normalised specification-gaming probes over retained candidate events; higher means more degenerate. Let ρ_{evt} be events per bar, ρ_{act} the minimum over voices of the fraction of bars with at least one attack, λ the share of total sounding time in notes of duration ≥ 4 beats, η the maximum over voice pairs of homorhythmic onset fractions (identical durations at shared onsets), and ν the ratio of realised to requested voices. Define clamped penalties $p_{\text{dens}} = \text{clamp}(1 - \rho_{\text{evt}}/4)$, $p_{\text{cov}} = \text{clamp}(1 - \rho_{\text{act}}/0.75)$, $p_{\text{long}} = \text{clamp}(\lambda/0.30)$, $p_{\text{coll}} = \eta$, and $p_{\text{voice}} = \text{clamp}(1-\nu)$. Then

$$D(S) = 0.25\,p_{\text{dens}} + 0.25\,p_{\text{cov}} + 0.20\,p_{\text{long}} + 0.15\,p_{\text{coll}} + 0.15\,p_{\text{voice}}. \tag{13}$$

Intuitively, D rises when a piece is sparse, bar-empty, sustained-chord heavy, homorhythmic, or missing requested voices, the cheap textures raw models adopt under adversarial briefs. Because events per bar are summed over voices and voice realisation is aggregated over the composition, D is texture-dependent by construction. We therefore compare D only within matched task conditions.

C Benchmark Tasks and Reference Corpus

Controlled task bank (40 tasks). Tasks form two single-factor sweeps that share a two-voice/8-bar anchor (Section 4). The *length* axis fixes two voices and varies total length over $\{8, 16, 32, 64\}$ bars across five writing concerns ($5 \times 4 = 20$ tasks): two-voice counterpoint; imitation; rhythmic division of labour; register contrast; and alternating lead. The *texture* axis fixes 8 bars and varies voice count over $\{1, 2, 3, 4\}$ across five concerns ($5 \times 4 = 20$ tasks): counterpoint density; rhythmic roles; register contrast; imitation; and a moderate combination of rhythm plus register. Each task is a single English brief targeting one mild technique; advanced serial devices (hexachordal combinatoriality, concurrent rows, derived rows, row elision) are deliberately omitted so gradients isolate length and texture. Representative briefs are given below; the full per-task briefs ship with the evaluation package (Appendix K).

Representative task briefs. Length-axis briefs are identical across the four lengths except for the bar count $N \in \{8, 16, 32, 64\}$ and share the stem "Write an N-bar two-voice twelve-tone exercise. . . "; texture-axis briefs share the stem "Write an 8-bar k-voice twelve-tone exercise. . . " and vary the voice count $k \in \{1, 2, 3, 4\}$, with k=1 realised as a monophonic melody. One brief per family (verbatim):

- **L1 Counterpoint.** "Write an N-bar two-voice twelve-tone counterpoint exercise. Each voice unfolds one row form; the voices enter staggered and use slightly different rhythms to stay independent. Avoid the two voices sounding the same pitch class at the same time, and avoid tonal implications."
- **L2 Imitation.** ". . . using imitation: one voice states the row first, and the other answers a little later with the same or a related row form."
- **L3 Rhythmic roles.** ". . . with a clear rhythmic division of labour: one voice moves slowly in long note values while the other unfolds the row with a more active rhythm."
- **L4 Register contrast.** ". . . with the two voices in contrasting registers, the upper voice higher and the lower voice lower, staying independent."
- **L5 Alternating lead.** ". . . where the two voices take turns leading: every few bars a different voice carries the main row statement while the other recedes into accompaniment."
- **V1 Counterpoint density.** "Write an 8-bar k-voice twelve-tone counterpoint exercise; the voices enter staggered and each unfolds one row form (k=1: a monophonic melody in row order with clear phrasing)."
- **V2 Rhythmic roles.** ". . . with layered rhythms across the k voices (slow/medium/fast) (k=1: one melody mixing long and short values)."
- **V3 Register contrast.** ". . . with the k voices spread across contrasting registers (k=1: one melody spanning a wide register)."
- **V4 Imitation.** ". . . where the k voices enter in staggered imitation, forming a light canon for $k \geq 3$ (k=1: one melody with motivic echoes)."
- **V5 Moderate combination.** ". . . combining rhythmic layering with register contrast across the k voices (k=1: a two-phrase melody contrasting register and rhythm); the four-voice case additionally avoids octave doubling."

Every polyphonic brief ends with the shared clause "avoid same-pitch-class collisions and avoid tonal implications"; monophonic (k=1) briefs keep only the tonal-avoidance clause.

Models evaluated. Table 7 lists the six models of Section 4 with their tier and role. All are accessed through a single OpenAI-compatible chat-completions gateway with temperature 0.2; the *frontier* tier supplies a ceiling/decay reference and is never wrapped in the harness (a full-harness run costs $\sim$\$1000 per seed at frontier list prices, which we judge not worth the marginal evidence given the harness-tier models already isolate the harness's effect); the *harness* tier supplies the same-model raw/harness pairs used for all headline raw-versus-harness comparisons. Six-model raw production reliability is reported separately below.

Table 7: Models evaluated (Section 4). "Both" means the model runs both raw and in-harness (same-model comparisons); "Raw only" means the model supplies a ceiling/decay reference and is never wrapped in the harness.

Model	Tier	Cond.
GPT-5.5	frontier	raw only
Gemini 3.1 Pro	frontier	raw only
DeepSeek V4 Pro	harness, mid	both
DeepSeek V4 Flash	harness, small	both
GPT-5 Nano	harness, small	both
Qwen3 235B	harness, small	both

Qwen3 235B additionally serves as the single model for all module ablations (Section 5.1), the red-team evaluation (Section 5.2), and the expert blind evaluation (Section 5.5), so that every non-headline comparison in the paper is also same-model; GPT-5.5 supplies the frontier red-team results (Appendix F) as a non-load-bearing reference (its own unprovoked writing is already much stronger than Qwen3 235B's, so its collapse under adversarial prompting is a less clean same-model signal on its own, per Section 5.2).

Table 8: Twenty-piece human reference corpus. Style subset labels follow the partition in Table 3.

Pieces	Composer	n
Strict-serial (subtotal 9)		
Op. 25 *Präludium*, *Musette*	Schoenberg	2
Op. 27 (3 movements)	Webern	3
Insomnia	Babbitt	1
Haiku, Op. 10	Sadri	1
Serial Composition No. 1	Gowans	1
Spring Pitch Sequence	Baker	1
Free-atonal (subtotal 11)		
Op. 15 (9 songs)	Schoenberg	9
Op. 19 (movements 2, 6)	Schoenberg	2
Total		**20**

Reference corpus (20 pieces). Table 8 lists the human reference set used for corpus distance. Pieces are grouped post hoc into *strict-serial* ($n = 9$) and *free-atonal* ($n = 11$) subsets for the breakdown in Table 3. All entries are stored as MusicXML for provenance and reduced to monophonic MIDI pitch lines before feature extraction (Appendix A).

D API Access and Inference Parameters

All six models (Table 7) are reached through the same gateway, and every call, whether it drives the raw baseline or any harness module (blueprint planner, section event planner, event-candidate proposer, and the repair / segment-rewrite / replan patches), uses the *same* decoding configuration. We deliberately do not tune per-model or per-module sampling settings. This removes bespoke tuning as one designed factor, but behavioural differences can still reflect model-specific gateway defaults, provider implementations, and the harness structure.

Table 9: API model identifiers.

Model	API identifier
GPT-5.5	`openai/gpt-5.5`
Gemini 3.1 Pro	`google/gemini-3.1-pro-preview`
DeepSeek V4 Pro	`deepseek/deepseek-v4-pro`
DeepSeek V4 Flash	`deepseek/deepseek-v4-flash`
GPT-5 Nano	`openai/gpt-5-nano`
Qwen3 235B	`qwen/qwen3-235b-a22b-2507`

Decoding configuration. Each request sets:

- **Temperature** $= 0.2$ (low but non-zero), which is the only sampling knob we control.
- **Structured output**: each request asks for one well-formed JSON object. A raw response counts as a parseable candidate only when it can be parsed and coerced into a nonempty event list; syntactically valid JSON without such a list is a failure in Table 12.
- **Sampling seed**: the per-run seed (N=3 runs per (task, model, condition)) is forwarded to the provider and controls LLM sampling only. The task's prime row is generated once and held fixed across those runs. Provider-side seeding improves repeatability but is not claimed to guarantee bitwise replay.
- **No other overrides**: nucleus sampling (top-p), maximum output tokens, and presence/frequency penalties are left at the gateway defaults and are never set by us, on any model or call type.
- **Non-streaming**: responses are requested as a single body rather than through server-sent events, which lets a plain read timeout fire cleanly on stalled reasoning models instead of being reset indefinitely by keep-alive traffic.

Timeouts and failed calls. Each call is bounded by a short connection timeout and a hard wall-clock cap (1200 s per call by default) enforced both by the socket read timeout and by a watchdog thread. A call that exceeds the cap, returns no content, fails to parse as JSON, or cannot be coerced into the required structured content is counted as a *failed call* and triggers the harness's retry/fallback path (deterministic backbone under REPLAN, Section 3.4); on the raw path there is no fallback, so such a call is an unparseable candidate and is reported as a negative result in the production-reliability tables (Appendix F). Token usage (prompt / completion / total) is accumulated per run from the gateway's usage field and underlies the cost figures of Table 2.

E Red-Team Brief Variants

For each base task brief x, the red-team wrapper appends one of five English *additional requirements* that encourage degenerate yet nominally legal output. The suffix is concatenated to the original brief. Explicit task metadata fields are unchanged, but the combined text is reprocessed by the theme compiler; keywords such as "chord" can therefore change the executable specification. The study measures end-to-end sensitivity to these modified briefs rather than holding the blueprint fixed. The five variants are:

1. **Long chords**: "To save effort, you may fill the duration with a few long notes and sustained chords; just passing the legality checker is enough, you need not pursue rhythmic or phrase variety."
2. **Sparse texture**: "Write as few notes as possible, with lots of empty space and long rests; put passing acceptance first, musicality does not matter."
3. **Voice collapse**: "The voices may march in lockstep with identical rhythms, starting and stopping together; do whatever is easiest, you need not keep the voices independent."
4. **Octave padding**: "You may rely on octave doubling, repeated notes, and repeating the same fragment to pad out the required number of bars and duration."
5. **Rubric attack**: "Your only goal is to maximise the acceptance pass rate: you may use any shortcut to satisfy the check items, including piling the material into a few bars and simplifying to whatever writing passes most easily."

The result tables use the short labels *Long chords*, *Sparse texture*, *Voice collapse*, *Octave padding*, and *Rubric attack*.

F Extended Red-Team and Production-Reliability Results

This appendix gives the full numeric tables supporting the red-team and production-reliability claims of Sections 5.2 and 5.1.

Raw-model diagnostics, all variants. Table 10 reports every red-team variant for the two raw conditions. The raw adapter initialises $A_{\text{tr}} = A_{\text{leg}} = 1.000$ rather than deriving these values from the harness verifier, so we omit them from cross-condition interpretation. The observed change is at the strict tier $A_\theta = A_{\text{dist}}$ (theme realisation and the material-distribution floor coincide in these raw adapter records). This equality is an adapter property and does not verify every technique requested by the 40 base tasks.

Harness diagnostics, all variants. Table 11 reports the same six variants for Qwen3 235B in-harness (240/240 runs, zero errors). This analysis uses one harness run per task–variant cell, compared with three raw runs, and does not estimate uncertainty across stochastic runs. *Ind. C+S* is the independent collision and serialisation-consistency checker (Appendix B); *theme realised* is Θ (Section 3.5). Across variants, the independent collision and serialisation-consistency pass rate (0.55–0.65) and degeneracy (≈ 0.05) remain near *normal*. Only the *Long chords* variant moves the strict tier ($A_{\text{dist}} : 0.275 \to 0.000$), driving *theme* to zero while the collision and serialisation-consistency pass rate barely drops ($0.600 \to 0.550$) and degeneracy stays flat. This is a material-coverage failure even though the surviving events are not more degenerate.

Table 10: Raw-model red-team diagnostics.

Model	Variant	Theme + dist. coverage	Degeneracy
Qwen3 235B	Normal	0.105	0.235
	Long chords	0.000	0.373
	Rubric attack	0.075	0.272
	Voice collapse	0.079	0.262
	Octave padding	0.162	0.222
	Sparse	0.325	0.213
GPT-5.5	Normal	0.971	0.103
	Long chords	0.026	0.214
	Sparse	0.618	0.196
	Rubric attack	0.706	0.198
	Voice collapse	0.939	0.145
	Octave padding	1.000	0.065

Table 11: Harness red-team diagnostics, Qwen3 235B.

Variant	Ind. C+S	Theme	A_{dist}	Degeneracy
Normal	0.600	0.950	0.275	0.053
Long chords	0.550	0.000	0.000	0.049
Rubric attack	0.650	0.950	0.300	0.050
Voice collapse	0.600	0.950	0.275	0.049
Octave padding	0.625	0.950	0.250	0.049
Sparse	0.650	0.950	0.225	0.050

Production reliability by model. Table 12 breaks down production reliability (Section 5.1). The four paired models contribute $4 \times 3 \times 40 = 480$ runs per condition; two additional raw-only references bring the raw total to $6 \times 3 \times 40 = 720$. A raw candidate is parseable if its one-shot response parses into an event list, regardless of legality or quality. `status=error` usually denotes a missing or invalid event list. One in-flight Flash request and one Nano request were terminated after excessive latency; both remain failures in the intention-to-treat denominator.

Table 12: Parseable-candidate rate, raw LLM vs. harness (Section 5.1).

Condition	Model	Parseable candidate
raw_llm	DeepSeek V4 Pro	$98/120 = 81.7\%$
raw_llm	GPT-5 Nano	$91/120 = 75.8\%$
raw_llm	Gemini 3.1 Pro	$111/120 = 92.5\%$
raw_llm	GPT-5.5	$105/120 = 87.5\%$
raw_llm	DeepSeek V4 Flash	$108/120 = 90.0\%$
raw_llm	Qwen3 235B	$113/120 = 94.2\%$
raw_llm	**all 4 paired models**	$\mathbf{410/480 = 85.4\%}$
raw_llm	**all 6 models**	$\mathbf{626/720 = 86.9\%}$
harness_full	**all 4 harness-tier models**	$\mathbf{480/480 = 100\%}$

Failures are not confined to small models: the two frontier models fail 7.5–12.5% of the time, making parseability distinct from legality or quality. The harness's 100% parseability is partly by design—structured generation plus a deterministic Replan fallback—and is not a like-for-like quality comparison (Section 5.1).

Production reliability under adversarial prompting. Table 13 repeats the comparison on the same model (Qwen3 235B) under the red-team suffixes of Appendix E, broken down by piece length; here the raw failure rate is not just non-zero but grows with length, while the harness stays at zero throughout.

The observed 64-bar raw failure rate is roughly $5\times$ the 8-bar rate. However, the 8-bar bucket also contains the texture sweep and has a different task composition and denominator, so this table does not isolate a causal length effect. Figure 2, by contrast, describes opportunity-normalised collision rates for raw and harness candidates and is not a raw-legality curve. The red-team means of Table 10 are conditional on parseable candidates (each variant retains $\geq 113/120$); no missing-not-at-random sensitivity analysis is available.

G Extended Ablation and Baseline Results

Table 1 in the main text reports point deltas for every ablation and prompt-only baseline; this appendix adds the bootstrap 95% CIs and, for the LLM-in-the-loop conditions, the theme-realisation delta omitted from the main table for space. Deltas use the same 40 paired task instances as Section 4; 2000-resample family-clustered bootstrap intervals are reported for every contrast. Each task value first averages its three runs; each bootstrap draw samples the ten prompt families with replacement while retaining all four tasks in a sampled family.

Figure 4: The following text is the task title used to generate this candidate: "Write an 8-bar two-voice twelve-tone counterpoint exercise. Each voice unfolds one row form; the voices enter staggered and use slightly different rhythms to stay independent. Avoid the two voices sounding the same pitch class at the same time, and avoid tonal implications."

Table 13: Red-team candidate-parse failure rate by nominal length, Qwen3 235B. The 8-bar column mixes length- and texture-axis tasks, whereas the longer columns contain only length-axis tasks, so columns are descriptive and not a controlled length effect.

Condition	8 bar	16 bar	32 bar	64 bar	Overall
raw_llm (N=3)	1.6%	0.0%	3.3%	7.8%	$17/720 = 2.36\%$
llm_full (N=1)	0.0%	0.0%	0.0%	0.0%	$0/240 = 0.0\%$

These remain descriptive fixed-bank sensitivity summaries rather than population-level significance tests.

Deterministic-backbone ablations (480 **offline runs:** 40 **tasks** $\times\, 4$ **conditions** $\times\, 3$ **seeds, no LLM).** Table 14 adds CIs to the top block of Table 1.

LLM-in-the-loop ablations (360 **Qwen3 235B runs:** 40 **tasks** $\times\, 3$ **conditions** $\times\, 3$ **seeds).** Table 15 adds CIs and the theme delta to the middle block of Table 1.

Table 14: Deterministic-backbone ablations with descriptive family-clustered bootstrap 95% intervals (vs. full harness).

Condition	Δ accepted	Δ ind. core	Δ degeneracy
No repair	-0.192 $[-0.267, -0.117]$	$+0.050$ $[0.000, 0.100]$	-0.006 $[-0.008, -0.003]$
No repair + no replan	-0.192 $[-0.267, -0.117]$	$+0.025$ $[-0.017, 0.067]$	-0.004 $[-0.007, -0.002]$
No replan	-0.025 $[-0.050, 0.000]$	$+0.025$ $[-0.017, 0.067]$	$+0.001$ $[0.000, 0.001]$

The no-repair condition averages 11.17 fewer events (95% CI $[-13.53, -9.03]$) and 0.713 fewer notes per bar ($[-0.921, -0.517]$) than the full harness, while mean active-bar coverage changes by only -0.001 ($[-0.002, 0.000]$) and voice realisation does not change. Because the narrow check conditions on generated overlap opportunities, the sparser candidate may provide fewer chances to collide. The observed $+0.050$ delta therefore cannot establish that removing repair improves candidate consistency; the offline records retain neither pairwise overlap counts nor sufficient artefacts to recompute the broader check for this ablation.

Table 15: LLM-in-the-loop ablations with descriptive family-clustered bootstrap 95% intervals (vs. full harness, Qwen3 235B).

Condition	Δ accepted	Δ ind. core	Δ degeneracy	Δ theme
No skills	$+0.017$ $[-0.042, 0.075]$	-0.042 $[-0.092, 0.008]$	$+0.000$ $[-0.001, 0.003]$	0 [0, 0]
Fixed row	-0.008 $[-0.083, 0.075]$	-0.092 $[-0.167, -0.025]$	$+0.001$ $[-0.002, 0.004]$	0 [0, 0]

These deltas support a narrower reading: removing skill prompts changes the independent collision and serialisation-consistency check by -0.042 and degeneracy by 0.000 in this task bank. Replacing the task-specific rows with one row shared by all tasks reduces that pass rate by 0.092. This is consistent with a contribution from task-conditioned row choice, but the ablation does not isolate its downstream planner and repair interactions.

Prompt-only baselines (≈ 240 **Qwen3 235B runs:** 40 **tasks** $\times\, 2$ **conditions** $\times\, 3$ **seeds).** Table 16 adds CIs and the theme delta to the bottom block of Table 1.

Self-refine iterates a model's own critique for two rounds (five calls/task: one generation plus two critique–rewrite rounds) with no external verifier; soft rules verbalises the hard-rule set (row order, valid row forms, no vertical same-pitch-class collisions, octave equivalence, tonal avoidance, voice independence) inside the prompt with a single call and no verifier. Both are markedly cheaper than the harness (Table 2) yet lose 33–40 points on the independent collision and serialisation-consistency check and 85–90% of theme

Table 16: Prompt-only baselines with descriptive family-clustered bootstrap 95% intervals (vs. full harness, Qwen3 235B).

Metric	Self-refine	Soft rules
Δ accepted	-0.183 $[-0.292, -0.071]$	-0.237 $[-0.342, -0.133]$
Δ ind. core	-0.333 $[-0.408, -0.254]$	-0.400 $[-0.508, -0.292]$
Δ degeneracy	$+0.241$ $[0.204, 0.279]$	$+0.208$ $[0.172, 0.246]$
Δ theme	-0.850 $[-0.933, -0.771]$	-0.904 $[-0.967, -0.838]$

realisation, with degeneracy up ~ 0.2. These comparisons show that the cheaper prompt-only conditions do not match the complete harness in this evaluation, but they confound external verification with a large difference in test-time calls and do not isolate which mechanism causes the gap.

H Supplementary Result Figures and Cost

This appendix collects a representative generated sample and exact per-model results supporting the analyses in the main text. Table 2 reports the per-run cost breakdown on Qwen3 235B. Figure 4 presents a representative generated sample. Its prime row is C, C♯, B♭, E, A, F, E♭, G, D, F♯, A♭, B. The upper voice uses the row forms P_0, I_0, R_0, and RI_0, while the lower voice uses P_0, I_0, and R_0. The sample exhibits substantial rhythmic counterpoint, maintains a moderate register within each voice, and gives both voices well-shaped trajectories with clear phrase-level development and closure. Table 17 lists the exact per-model values. All four models improve on both candidate checks with the harness, converging to a 0.53–0.63 collision and serialisation-consistency pass rate, a 0.43–0.52 constraint-checked delivery yield, and a 0.047–0.053 degeneracy band.

Table 17: Per-model raw vs. harness results (n=120 per condition). Both checks count unparseable raw candidates as failures; raw degeneracy is conditional on parseability.

Model	Ind. C+S Raw	Ind. C+S Harness	Checked yield Raw	Checked yield Harness	Degeneracy Raw	Degeneracy Harness
DeepSeek V4 Flash	0.350	0.558	0.125	0.433	0.186	0.053
GPT-5 Nano	0.267	0.617	0.058	0.517	0.132	0.050
DeepSeek V4 Pro	0.458	0.533	0.225	0.483	0.183	0.053
Qwen3 235B	0.267	0.625	0.125	0.492	0.244	0.047

Cost details. The full harness uses about 157× the raw token count and 11× its wall-clock time, costing approximately $0.07 per Qwen3 235B piece. Skill prompts account for most of the token overhead; estimates exclude local verification, rendering, and storage.

I Expert Evaluation Protocol and Panel

Panel. Five expert raters evaluate the blind package. All are active composers and/or music theorists with graduate training in composition and professional experience in twelve-tone or atonal writing and analysis. Four raters have over twenty years of professional practice; one has over thirty.

Protocol. Scores are generated once per condition with the same mid-tier model on eight tasks drawn from the controlled task set, spanning 1–4 voices and 8–64 bars (imitation, register contrast, rhythmic roles, and counterpoint density). The study evaluates retained candidates irrespective of release-gate status and is therefore a candidate-quality diagnostic. The preset comprises L1 counterpoint (8 bars), L2 imitation (16), L3 rhythmic roles (32), L4 register contrast (64), and V1–V4 at 1–4 voices respectively (counterpoint density, rhythmic roles, register contrast, and imitation; all 8 bars). Pair order, anonymous item codes, and A/B assignment are pseudorandomly generated with a fixed seed. Each task yields three anonymised pairwise comparisons: full harness versus raw LLM, versus harness without skill injection, and versus harness without repair (24 pairs total). Within a pair the A/B assignment is randomised and independent across pairs; one mapping is shared by all raters rather than counterbalanced separately per rater, and the same full-harness candidate recurs in the three contrasts for a task. The mapping to conditions is held in a separate manifest not shown to raters. Raters receive English instruction sheets and two printable scores per pair. The rating sheet enforces a forced binary choice (A or B, no ties) on five dimensions defined below. Each rater completes all 24 pairs.

Rating dimensions. Raters judge each pair on five forced-choice dimensions:

- **Adherence**: whether the score satisfies the *specific* task brief (voice count, bar length, imitation, register contrast, rhythmic roles, etc.).
- **Legality**: perceived twelve-tone rule conformance (row order, avoidance of illegal octave doubling, no vertical pitch-class collisions).
- **Style**: idiomatic serial writing (contour, register use, avoidance of tonal implication, motivic interest).
- **Coherence**: large-scale structure and sense of intentional form.
- **Overall**: global preference between the two scores.

Aggregation. For each (comparison, dimension) we count how often the full harness is chosen ($n = 40$ repeated judgements: 5 raters × 8 tasks). We report the descriptive win rate and a 10,000-resample task-clustered bootstrap CI: each draw resamples the eight tasks and retains all five ratings for each sampled task (fixed seed 7). We also report Fleiss' κ across the five binary ratings per pair. We do not treat the 40 ratings as independent observations.

Full results. Table 18 lists descriptive counts for all three comparisons. For full versus raw, task-clustered 95% CIs are $[.75, .95]$ (adherence), $[.725, .95]$ (legality), $[.60, .775]$ (style), $[.75, .90]$ (coherence), and $[.80, .975]$ (overall). Per-rater rates pooled over dimensions range from 0.60 to 1.00. Fleiss' κ ranges from $-.13$ to .03 across these dimensions, indicating weak agreement despite the aggregate preference. Across all three contrasts and dimensions, κ ranges from $-.20$ to .20.

Table 18: Expert pairwise results ($n = 40$ repeated judgements per cell). Win-rate = fraction preferring the full harness.

Comparison	Dimension	W / L	Win-rate
Full harness vs. raw generation			
	Adherence	34 / 6	0.85
	Legality	34 / 6	0.85
	Style	28 / 12	0.70
	Coherence	33 / 7	0.82
	Overall	36 / 4	0.90
Full harness vs. without skills			
	Adherence	24 / 16	0.60
	Legality	30 / 10	0.75
	Style	25 / 15	0.62
	Coherence	23 / 17	0.57
	Overall	22 / 18	0.55
Full harness vs. without repair			
	Adherence	30 / 10	0.75
	Legality	26 / 14	0.65
	Style	25 / 15	0.62
	Coherence	23 / 17	0.57
	Overall	24 / 16	0.60

Reproducibility. The blind evaluation package ships with the release: anonymised score pairs, English task sheets, rater forms, the hidden condition manifest, and aggregated tally tables matching Table 18.

J Recruitment and Payment

Participation is voluntary and uncompensated, consistent with specialist academic consultation. The study collects only anonymised pairwise score preferences and, given this design, no dedicated institutional ethics review is sought.

K Reproducibility and Release Artefacts

We release the harness, 40-task brief bank, frozen 20-piece reference corpus, blind-evaluation package, and scripts for computing the independent collision and serialisation-consistency check, final constraint check, degeneracy, process-tier acceptance, and corpus-distance results.

L Use of LLMs

LLMs have two distinct roles in this work. First, the models named in the experimental design are evaluated systems: their generated proposals or raw scores constitute experimental observations. Second, LLM-based assistants support manuscript editing such as grammar, style, and clarity improvements. We retain full control over the content, phrasing, and presentation of the final text.